

Unit 5: Geographic Pattern Analysis

5.1 Distance Measurement

In GIS, distance measurement is vital for understanding how geographic features relate to each other in space. It helps in navigation, planning, and spatial analysis. Distances can be analyzed in three main ways:

1. Absolute Distance

The shortest, straight-line distance between two geographic points, also called Euclidean distance. This is a purely geometric measure that doesn't consider any physical obstacles (like rivers or buildings) or network constraints (like roads or traffic).

- **Characteristics:**
 - Measured using fixed units such as meters, kilometers, or miles.
 - Assumes a flat surface unless elevation is considered.
- **Example:** The direct line distance ("as the crow flies") between Kathmandu and Pokhara.

2. Relative Distance

The distance expressed in terms of travel time, cost, or effort required to move between two locations, rather than physical units. Takes into account the real-world conditions like road types, terrain, traffic congestion, or transportation mode.

- **Characteristics:**
 - More realistic for human and vehicle movement.
 - Variable depending on circumstances (e.g., traffic or weather).
- **Example:** It may take 30 minutes to drive 10 km in heavy traffic, or 15 minutes in light traffic — though the absolute distance remains the same.

3. Functional Distance

A measure of interaction, flow, or connectivity between two places, regardless of physical proximity. Reflects the intensity of social, economic, or cultural links between areas. Places can be functionally "close" due to strong interactions even if they are far apart.

- **Characteristics:**
 - Not measured in physical or temporal units.

- Common in studies involving migration, trade, service access, and communication.
- **Example:** Two cities engaged in daily business trade are functionally close, even if separated by 500 km.

5.2 Statistical Surfaces

In GIS, statistical surfaces represent continuous spatial phenomena where values vary smoothly across space (e.g., elevation, rainfall, pollution). These surfaces are typically created and analyzed using raster data or interpolated surfaces.

1. Characteristics of Statistical Surfaces

- Represent **continuous data** (every location has a measurable value).
- Modeled as **grids** or **TINs (Triangulated Irregular Networks)**.
- Used in **environmental, meteorological, hydrological, and terrain analysis**.
- Examples: **elevation surface, temperature distribution, soil moisture**, etc.

2. Working with Surface Data

- Surface data can come from:
 - **Raster layers** (e.g., Digital Elevation Models - DEMs).
 - **Point datasets** (e.g., weather stations) converted into surfaces using interpolation.
- Tools in QGIS/ArcGIS allow:
 - **Extraction** of surface values.
 - **Overlay** of surface and vector data.
 - **Surface profiling**, slope, aspect, and **visibility analysis**.

3. Predicting Values with Interpolation

Interpolation estimates unknown values at unsampled locations based on known data points.

Common Interpolation Techniques:

- **IDW (Inverse Distance Weighting):**
 - Closer points have more influence on prediction.
- **Kriging:**
 - Advanced geostatistical method considering spatial autocorrelation.
- **Spline:**

- Smooth surface fitting through known points.

Applications:

- Creating temperature or rainfall maps from station data.
- Estimating air pollution levels across a city.
- Predicting soil pH in agricultural planning.

5.3 Topological Surfaces

In GIS, topological surfaces refer to representations of spatial relationships among features such as adjacency, connectivity, and containment. Unlike statistical surfaces, which focus on values over a surface, topological surfaces focus on how features are linked or structured in space, often within vector data.

1. Characteristics of Topological Surfaces

- **Topology** describes how features share geometry (edges, vertices).
- Focus is on **relationships** rather than coordinate accuracy.
- Maintains data **integrity** during editing or analysis (e.g., no gaps or overlaps in adjacent polygons).

2. Components of Topological Surfaces

- **Nodes**: Start/end points of lines.
- **Edges (arcs)**: Define borders of polygons or connections between nodes.
- **Polygons**: Areas formed by closed edges.
- **Faces**: Surface parts defined by edges and nodes.

3. Importance of Topology in GIS

- **Detects errors** like gaps or overlaps in maps (e.g., land parcels).
- Enables **network analysis**, such as routing and connectivity.
- Supports **overlay operations** (e.g., intersect, union).
- Ensures **data quality** in cadastral (land record) systems.

4. Examples of Topological Relationships

- A river **connects to** a lake (connectivity).
- Two countries **share a border** (adjacency).
- A building is **within** a parcel boundary (containment).

5.4 Networks in GIS

A network in GIS refers to a system of interconnected linear features (like roads, rivers, or utility lines) used to model movement or flow. Network analysis helps in solving real-world problems such as finding the shortest path, optimizing travel routes, and directing traffic flow.

1. Connectivity Measurement

Determines how well different nodes (points) or segments in a network are connected.

- **Use:** Helps identify isolated areas or disconnected parts of the network.
- **Example:** Ensuring all intersections in a road map are reachable from each other.

2. Impedance Values

Represents the cost of traveling through a segment in the network (e.g., time, distance, fuel).

- **Use:** Influences route choice in shortest path or least-cost analysis.
- **Example:** A busy highway may have a higher impedance due to congestion.

3. One-Way Paths

Road segments that allow travel in only one direction.

- **Use:** Ensures routes comply with real-world traffic rules.
- **Example:** Directional restrictions on urban streets.

4. Circuits

A closed path where the start and end points are the same.

- **Use:** Useful in delivery routing or inspection tours.
- **Example:** Postal delivery route returning to the depot.

5. Turns and Intersections

- **Turns:** Include turn restrictions (e.g., no left turn) and turn costs (e.g., time delays).
- **Intersections:** Points where two or more paths meet; often modeled as nodes.
- **Use:** Essential for realistic routing and navigation.

6. Directing Traffic and Exploiting Networks

Uses the network's structure and rules to manage and guide movement.

- **Use:**
 - Traffic flow optimization
 - Emergency response routing
 - Utility line tracing
- **Tools:** QGIS Road Graph plugin, pgRouting with PostGIS

Example Applications

- Emergency vehicle routing
- Public transport network analysis
- Delivery and logistics planning